

- 1) NLP is a branch of AI that enables computers to understand human language. It is used to ~~process~~ analyze and generate text or speech. 
- 2) The main goal are to understand language, extract meaning and enable effective human computer communication.
- 3) NLP is applied in chatbots, machine translation, sentiment analysis, speech recognition, summarization and information retrieval.
- 4) NLP is the overall field, NLU focuses on understanding languages and NLG focus on generating human like languages.
- 5) Major challenges include ambiguity, context understanding, sarcasm, slang, Multiple language interpretations.
- 6) Human language is different because it contains ambiguity, context, idioms, slang and multiple possible interpretation.
- 7) A ambiguity occurs when a sentence has more than one possible interpretation.
- 8) Lexical concerns word meaning, syntactic concerns structure, semantic concerns sentence meaning and pragmatic concerns context.

- 9) A language model predicts or generates a sequence of words and helps generate meaningful language.
- 10) Rule-based NLP uses manually linguistic rules, while statistical NLP learns patterns and probabilities from data.
- 11) A regular Expression is a pattern used to search, match, extract or manipulate text.
- 12) Common symbols include $*$, $+$, $?$, $.$, $^$, $\$$, and character classes such as $[]$.
- 13) $*$ Means zero or more occurrences of the preceding character or pattern.
- 14) $+$ Means one or more occurrences of the preceding character or pattern.
- 15) Character classes define a set or range of characters that $*$ allows zero or more occurrence, whereas $+$ requires at least one occurrence.
- 16) Character classes define a set or range of characters that can be matched in a text pattern.
- 17) An FSA is a computational model containing states and transitions that recognizes pattern valid input.
- 18) The main components are states, input symbols, transitions, an initial state and accepting states.

- analysis and recognizing language pattern.
- 20) A DFA has a one transition for each state input pair while an NFA can have multiple possible transitions.
- 21) Morphological is the study of the internal structure, for motion and modification of words.
- 22) A morpheme is the smallest linguistic unit that carries meaning or grammatical information.
- 23) A free morpheme can stand alone as a word, while a bound morpheme must attach to another morpheme.
- 24) Inflectional morphology changes a word's grammatical form without creating a new word.
- 25) Derivational morphology creates a new word by adding a prefix or suffix to an existing word.
- 26) Inflectional morphological expresses grammatical changes, while derivational morphological create new words or meanings.
- 27) "walk" changing to "walked" is an example of inflectional morphology.
- 28) "Teach" changing to "teacher" is an example of derivational morphology.
- 29) Morphological parsing divides a word into its morphemes and identifies their grammatical rules.

- 30) It effectively analyzes word structures and morphemes using finite state machines or transducers.
- 31) Stemming reducing words to a common stem by removing using finite state machines or transducers.
- 32) Stemming Lemmatization converts a word into its valid dictionary base form or lemma.
- 33) Stemming uses simple word-ending rules, while lemmatization uses linguistics knowledge to produce valid base words.
- 34) Porter stemmer is an algorithm that removes common English suffixes to reduce words to their stems.
- 35) It can produce non-words; it may over stem or under stem and does not consider word meaning or context.
- 36) An N-gram is a sequence of N consecutive words or tokens in a text.
- 37) A unigram model considers one word at a time and estimated individual word probabilities.
- 38) A bigram model considers two consecutive words to model their relationship.
- 39) A trigram model considers three consecutive words to capture local language effect.
- 40) Unigram uses one word, bigram uses two consecutive words and trigram uses three consecutive words.

n-gram probability is calculated as $P(w_2/w_1) = \text{const} \frac{\text{Count}(w_1/w_2)}{\text{Count}(w_1)}$

- 42) N-gram models capture local word relationships and are useful for prediction, speech recognition and text generation.
- 43) An unsmoothed N-gram model uses observed frequency directly, giving unseen N-grams a probability of zero.
- 44) Unseen N-grams receive zero probability, which can incorrectly make an entire sentence probability zero.
- 45) Smoothing assigns small non-zero probabilities to unseen N-grams.
- 46) Smoothing prevents zero probabilities and improves a language model's performance on unseen text.
- 47) Back off uses a lower-order N-gram model when the required higher-order N-gram is unavailable.
- 48) Deleted interpolation combines probabilities from different N-grams order using weighted interpolation.
- 49) Entropy is a measure of uncertainty or unpredictable in a language model.
- 50) High entropy indicates that the next word or sequence is different for the model to predict.
- 51) POS tagging assigns grammatical categories such as noun, verb, adjective and adverb to words.
- 52) POS tagging helps identify grammatical rules and support parsing, translation and information extraction.

- 53) Major English word classes include: nouns, verbs, adjectives, adverbs, determiners, prepositions and conjunctions.
- 54) A noun is a word that represents a person, place, thing or idea.
- 55) A pronoun is a word used instead of a noun or noun phrase.
- 56) A verb is a word that expresses an action, event or state.
- 57) An adjective is a word that describes or modifies a noun or pronoun.
- 58) An adverb is a word that modifies a verb, adjective or another adverb.
- 59) A determiner is a word that specifies or identifies a noun, such as "the", "a", or "this".
- 60) A preposition shows the relationship between a noun or pronoun and another word, phrase or clause in a sentence.
- 61) A conjunction connects words, phrases or clauses in a sentence.
- 62) A POS tag is a defined collection of labels used to represent different grammatical categories.
- 63) POS ambiguity occurs when the same words can receive different grammatical tags depending on context.
- 64) Rule-based POS tagging assigns grammatical tags using manually defined linguistic rules.

- stochastic POS tagging uses statistical probabilities learned from tagged text to select the most likely POS tag.
- 66) Transformation-based POS tagging starts with initial tags and applies correction rules to improve tagging accuracy.
 - 67) Rule-based tagging uses predefined rules, while stochastic tagging uses probabilities learned from training data.
 - 68) The same word can have different POS tags because its grammatical role changes according to sentence context.
 - 69) "Book" is a noun in "I read a book", but a verb in "Book a ticket".
 - 70) Context helps determine the correct grammatical tag when a word has multiple possible POS categories.
 - 71) A CFG is a formal grammar used to describe the syntactic structure of sentences.
 - 72) The four components are terminals, non-terminals, production rules, and a start symbol.
 - 73) A terminal symbol is a basic symbol that appears directly in the final generated sentence.
 - 74) A non-terminal symbol represents a grammatical category that can be expanded using production rules.

- 75) A production rule defines how one grammatical symbol can be replaced or expanded into other symbols.
- 76) A parse tree is hierarchical representation showing the grammatical structure and derivation of a sentence.
- 77) A parse tree shows grammar-based derivation, while a syntax tree emphasizes the syntactic relationships.
- 78) The rule $S \rightarrow NPVP$ means a sentence consists of a noun phrase followed by a verb phrase.
- 79) Syntactic analysis determines the grammatical structure and relationships among words in a sentence.
- 80) Parsing is the process of analyzing a sentence according to grammar to determine structure.
- 81) Bottom-up parsing starts with the input words and combines them to reach the grammar start symbol.
- 82) Top-down parsing moves from the start symbol to words while bottom-up parsing moves from words to the symbol.
- 83) Top down parsing starts from the grammar start symbol and works toward matching the input words.
- 84) An early parser is a general-purpose chart parsing algorithm used for context free grammar.

Early parsing is useful because it can effectively handle complex and ambiguous content, tree grammars.

- 86) Grammatical agreement means related words match in grammatical features such as number, person, gender.
- 87) Subject-verb agreement requires a verb to match its subject in number and person.
- 88) Subcategorization specifies the types of complements or arguments that a word especially a verb can take.
- 89) A feature structure represents linguistic information using attribute-value pairs such as number.
- 90) A PCFG is a CFG in which each production rule is assigned a probability.
- 91) Discourse refers to connected sentences that together form meaningful communication.
- 92) Discourse analysis studies the relationships, organization and meaning of sentences across a text.
- 93) Coreference resolution identifies the entity to which a word refers.
- 94) Anaphora resolution identifies a previous entity referred to by an expression such as "Ravi is tired because he worked".
- 95) Text coherence is the logical and meaningful connection among ideas and sentences to text.

- 96) Discourse structure describes how sentences or sections are organized and related through relationships.
- 97) Cohesion refers to grammatical and lexical links b/w sentences, while coherence refers to the overall logical flow of ideas.
- 98) A dialogue system is an NLP system that communicates with users through conversational interactions.
- 99) A conversational agent is software that understands user input and generates conversational response.
- 100) A dialogue act represents the function of an utterance, such as a question or request answer.